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Hidden Scars: An Analysis of Gender Health Inequalities in the Netherlands

Project 1 Research

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1. Introduction

In the affluent context of contemporary Western societies, the presence of significant health disparities, deeply rooted in socioeconomic inequalities, is often overlooked. Despite the advanced medical facilities and high standards of living, these disparities permeate daily life (Jenkinson, 2001).

Study Background

This thesis focuses on the Netherlands due to its paradoxical nature: high healthcare standards and substantial public health investment juxtaposed against persistent health outcome disparities among its population. This setting provides a prime case study for investigating how nuanced factors like gender, occupational conditions, marital status, age, and hours worked contribute to health inequalities in a well-developed nation (Bertens & Vonk, 2020).

Research Hypothesis

The central hypothesis of this study is that in the Netherlands, the general self-reported health status is significantly influenced by gender, with conditioning factors including work intensity, occupation, age, and marital status. This thesis seeks to elucidate the complex interplay among these variables and their impact on gender-specific health disparities.

Contributors

This research was undertaken by a proficient team from Corvinus University of Budapest, comprising Meshkovski Nikola, Yarovoy Daniil, Katona Balázs Levente, and Kákonyi Bence.

2. Literature Review

We seek to contextualize the research within the scope of existing studies. Winchester (2021) identifies significant gender health disparities in UK healthcare, highlighting systemic biases against women in treatment and care. Drawing on multiple studies and reports, it reveals a gender health gap with women often receiving inferior medical treatment, particularly in chronic conditions like dementia and maternal healthcare. The study emphasizes the critical need for healthcare reforms to address these disparities. Notably, Broom (2008) delivers a concise analysis of the interplay of gender and SES and their impact on health outcomes, with a focus on obesity and smoking. Published in the Australian Journal of Social Issues, her research utilizes epidemiological data and multivariate analysis to expose significant health

disparities linked to gender and socioeconomic status. She reveals that lower SES is associated with increased health risks, especially among marginalized groups, emphasizing the critical need for health policies to incorporate gender and socioeconomic factors to effectively address health inequalities. Furthermore, Artazcoz, L., et al. (2016) used ordered logistic regression to analyze data from the 2010 European Working Conditions Survey involving 30,851 workers, finding that long working hours negatively impact health, with significant variations across gender and different welfare regimes. Similarly, Čehovin Zajc, Jožica, and Kohont, Andrej (2017) analyzed data from a Slovenian survey of 1,082 employees and discovered a strong correlation between high work intensity and poor health outcomes, underscoring the role of work-related stress in health disparities.

3. THE DATA

DATA SOURCE

The primary data for this research comes from the 2013 Eurostat's Microdata Statistics on Income and Living Conditions (EU-SILC) (Eurostat, 2024), which offers a robust dataset for examining health disparities in the Netherlands. The EU-SILC dataset includes a wealth of information on variables like self-reported health conditions, gender, occupational status, marital status, age, and hours worked, making it highly relevant for our analysis. This dataset uses synthetic data to preserve respondent privacy while providing a representative sample of the Dutch population through a stratified sampling method. This stratification enhances the accuracy of the data and has been a trusted method in research for many years, allowing for generalizable and cost-effective insights into socioeconomic disparities (Stephan, 1941).

DATA SAMPLE

For our analysis, the initial sample included about 18,157 actively working individuals aged between 19 and 81 years, targeting those most likely to be influenced by the study factors. Following an extensive data cleaning and transformation process to ensure the data's quality and relevance to our research hypothesis, we refined the sample size down to 1,521 participants, excluding variables that did not directly relate to our specific research focus.

CONTENT OF VARIABLES

The thesis employs key variables to explore health disparities: "General Health" is rated from 1 (very good) to 5 (very bad); "Female" is recoded from gender for clarity, represented as 0 for not female and 1 for female; "Marital Status" is simplified to binary, indicating 0 for previously married and 1 for never married; "Occupation" is classified under ISCO-08, focusing on five major groups; "Hours Worked Weekly" is excluded due to non-compliance with the parallel line assumption; "Age" is derived directly from birth year, accurately reflecting each participant's age at the time of the survey. This streamlined variable framework supports a focused analysis of gender and other demographic impacts on self-reported health.

INITIAL DATA CLEANING

Records with missing data for key variables like general health, weekly hours worked, and marital status were excluded, significantly reducing the dataset size, these observations would have been ignored either way when conducting our ordinal logistic regression analysis.

STATISTICAL OVERVIEW

Most participants rated their health as 'good' (2), see Appendix Figure 1, with an average slightly above this. The dataset comprises roughly 52% females and 30% of participants have never been married. Occupational distribution among participants ranges from 18% to 24% per category. Participant ages span from 19 to 81 years, averaging around 49.

See Appendix Figure 2.

4. METHODOLOGY

ANALYTICAL TECHNIQUES

Ordinal Logistic Regression: Given the ordered nature of the health status variable, ordinal logistic regression is used to estimate the probability of each health category as a function of the independent variables.

Brant Test: This test is performed to verify the assumption of parallel lines, which is crucial for the appropriateness of using ordinal logistic regression.

Estimation of Marginal Effects: After fitting the regression model, marginal effects are calculated for each independent variable to interpret how a unit change in each variable affects the probability of reporting different levels of health status.

ESTIMATION EQUATION

The core of our analysis is based on an ordinal logistic regression model, formulated to understand the influences on general health status.

Model 1: Ordinal Logistic Estimation Equation

$$Logit\big(P(Y < j)\big) = \beta_0 + \beta_1 \cdot \text{ Female} + \beta_2 \cdot \text{ Marital Status} + \beta_3 \cdot i.Occupation + \beta_4 \cdot age + error$$

The "Female" variable is expected to show that women are more likely to report poorer health compared to men, possibly due to societal and biological differences, as well as greater health awareness. Marital status likely indicates better health for married individuals due to the benefits of social support and shared resources. Occupation affects health variably depending on job-related stress and working conditions, with different occupations having distinct health risks. Finally, age typically correlates with declining health, as older individuals are more prone to chronic conditions and physical deterioration. Together, these variables provide a comprehensive understanding of the factors influencing health disparities.

5. RESULTS

Table 1: Results from Conducting the Ordered Logistic Regression

General Health	Log Likelihood	P -Value
Female	0,38	0,00
Not Married	0,02	0,86
Occupation Teaching Professional (Base Outcome)		
Business Professional	0,12	0,45
Business Associate Professional	0,03	0,84
General and Keyboard Clerks	0,02	0,92
Sales Workers	0,11	0,47
Age	0,00	0,96

Source: Author's calculations based on data from the 2013 release of EU - SILC.

Table 1 reveals that the "Female" variable statistically significantly impacts health, with women more likely to report worse health than men. There are no significant effects on health outcomes from marital status, occupational categories, or age.

Table 2: Results from Conducting the Brant Test (having included "Hours Worked")

Variables	p>chi2
Female	0.445
Not Married	0.993
Teaching Prof. (Base Outcome)	
Business Prof.	0.632
Business Associate Prof.	0.884
General and Keyboard Clerks	0.803
Sales workers	0.298
Age	0.061
Hours Worked	0.009

Source: Author's calculations based on data from the 2013 release of EU - SILC.

Table 2 shows that the variable "Hours Worked" failed the parallel line assumption with a p-value smaller than required (0.05 > 0.009), indicating inconsistent effects across different health states, leading to its exclusion from further analysis. Despite this, our study maintains its robustness and focus, after reassessing and confirming that all other variables like gender, age, marital status, and occupation adhere to the parallel line assumption. This allows us to continue exploring gender-specific health disparities in the Dutch population effectively, emphasizing the impact of these retained variables on health outcomes.

Table 3: Results from Conducting the Estimation of Average Marginal Effects for Female

Female	Coefficient	Std. Error	P - Value
1 - Very Good	-0,061	0,016	0,000
2 - Good	-0,006	0,004	0,115
3 - Fair	0,052	0,014	0,000
4 - Bad	0,013	0,004	0,001
5 - Very Bad	0,003	0,001	0,013

Source: Author's calculations based on data from the 2013 release of EU - SILC.

In Table 3, our analysis shows that in the Netherlands, women are 6.1% less likely than men to report 'very good' health and more likely to report worse health states, 5.2% more likely to report 'fair', 1.3% more for 'bad', and 0.3% more for 'very bad'. The probability of reporting 'good' health is not statistically significant. Overall, these results suggest that women tend to rate their health lower than men.

The cross-tabulation results in Table 4 in the Appendix directly support the findings that women in the Netherlands report their health as worse compared to men. 70% of females versus 30% of males report their health as 'bad', with women consistently more likely to report worse health conditions across all categories. This discrepancy suggests that gender differences in health perception or reporting may influence these outcomes, highlighting the need for further research.

6. SUMMARY

This thesis critically explores the contrast between the Netherlands' high healthcare standards and its socio-economic health disparities, focusing particularly on gender's impact on self-reported health outcomes. Using 2013 Eurostat data and ordinal logistic regression, the study confirms that gender significantly affects health reporting, with men likely underreporting issues due to societal norms, potentially skewing health assessments. Key limitations include reliance on self-reported data, which may not fully reflect actual health statuses. Future research should probe societal influences on health reporting and validate findings with medical records to better understand gender-based health disparities.

7. References

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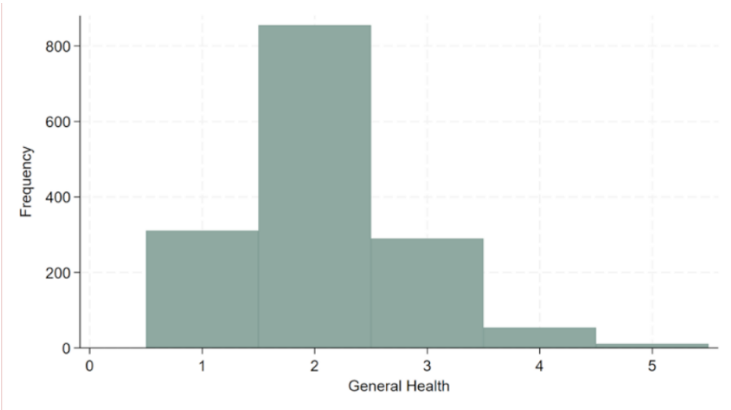
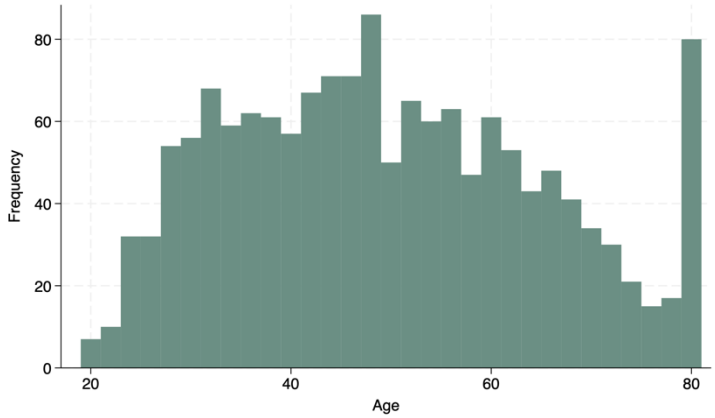
8. APPENDIX

Figure 1: Distribution of General Health Variable



Source: Author's calculations based on data from the 2013 release of EU - SILC.

Figure 2: Distribution of Age Variable



Source: Author's calculations based on data from the 2013 release of EU - SILC.

Table 4: Cross Tabulation of Gender and General Health Variables

General Health	Male	Female	Male in %	Female in %	Total
1	170	141	55%	45%	311
2	416	439	49%	51%	855
3	122	168	42%	58%	290
4	16	38	30%	70%	54
5	5	6	45%	55%	11
Total	729	792	48%	52%	1521

Source: Author's calculations based on data from the 2013 release of EU – SILC, Inspired by colleague Yarovoy Daniil.